

MIDTERM TEST, FALL 2024 Course: CS365 - Intro. to COA Duration: 60 minutes		Student name: Student Id: Order. Class Id:
Assessors' name and signature	Examiners' name and signature	Total score

Test Id: 1 (This test consists of 10 questions)

Note: *This is an open-book exam and students are allowed to use books, smartphones, etc.*

Questions with one correct answer

Question 1. What is the decimal value of this 8-bit two's complement number 1100 0101 after shifting the number right two bits?

- ☐ 15
 ☐ -49
 ☐ 49
 ☒ -15

Question 2. Suppose that register \$t2 contains 0000 0000 0000 0000 0000 1101 0000 0000 and register \$t1 contains 0000 0000 0000 0000 0011 1100 0000 0000. Which of the following MIPS instructions places the value 0000 0000 0000 0000 0000 1100 0000 0000 in register \$t0?

- ☐ and \$t1, \$t2, \$t0
 ☐ add \$t1, \$t2, \$t0
☒ and \$t0, \$t1, \$t2
 ☐ add \$t0, \$t1, \$t2

Question 3. Which of the following MIPS codes loads the 32-bit constant 0x00300100 into register \$s0?

- ☐ lui \$s0, 256
 ☒ lui \$s0, 48
 ☐ lui \$s0, 46
 ☐ lw \$s0, 256
☐ addi \$s0, \$s0, 48
 ☐ addi \$s0, \$s0, 256
 ☐ add \$s0, \$s0, 48

Question 4. Suppose that the computer A, which has a 400MHz clock, runs a program P in 10 seconds. The computer B, which has twice the clock rate of A, runs P in 6 seconds. What is the number of clock cycles required for P on B?

- ☐ 3600×10^6 cycles
 ☐ 4500×10^6 cycles
 ☐ 4600×10^6 cycles
 ☒ 4800×10^6 cycles

Question 5. What is the machine instruction corresponding to the MIPS instruction: add \$s0,\$a1,\$t7?

- ☐ 0x00BF8020
 ☐ 0x00AE8020
 ☒ 0x00AF8020
 ☐ 0x00AD8020

Question 6. Suppose that we have two computers A and B implementing the same instruction set architecture. Computer A has a clock cycle time of 1 ns and a CPI of 2.0 for a program P, and computer B has a clock cycle time of 1.5 ns and a CPI of 1.2 for P. How much faster is computer A than computer B for this program?

- ☐ 1.5 times
 ☐ 1.3 times
 ☐ 1.2 times
 ☒ 0.9 times

Constructed-response questions

Question 7. Assume that \$a0 initially contains 5 and \$v0 is used for the output. Add comments to the following MIPS code and describe in one sentence what it computes.

```

begin: addi $t0, $zero, 0 .....
      addi $t1, $zero, 1 .....
loop:  slt  $t2, $a0,  $t1 .....
      bne  $t2, $zero, fin .....
      add  $t0, $t0,  $t1 .....
      addi $t1, $t1,  2 .....
      j    loop .....
fin:   add  $v0, $t0,  $zero .....
.....
.....

```

Question 8. What is the MIPS assembly code corresponding to the following machine code?

```

100011010010100000000010010110000 .....
00000001001001000001000000000100000 .....
101011010010100000000010010110000 .....

```

Question 9. Compile the following C program into MIPS assembly code.

```

int func (int n)
{
    int i=0;
    while (i<n)
        i=i+1;
    return i;
}

```

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```

Question 10. Construct the full truth table described by the following Verilog module.

```

module func {A, B, C, S, D}; .....
    input A, B, C; .....
    output S, D; .....
    assign S=A&B&C; .....
    assign D=(A^B)|(B^C)|(C^A); .....
endmodule .....
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```

MIDTERM TEST, FALL 2024 Course: Intro. to Comp. Org. and Arch. Course Id: CS365, Duration: 60 minutes		Student name: Student Id: No. Class Id:
Assessors' name and signature	Examiners' name and signature	Total score

Test Id: 2 (This test consists of 10 questions)

Note: *This is an open-book exam and students are allowed to use books, smartphones, etc.*

Questions with one correct answer

Question 1. What is the decimal value of this 8-bit two's complement number 1110 0101 after shifting the number left two bits?

- ☐ -148
 ☐ 108
 ☐ 148
 ☒ -108

Question 2. Suppose that register \$t2 contains 0000 0000 0000 0000 1011 0000 0000 and register \$t1 contains 0000 0000 0000 0000 0011 1100 0000 0000. Which of the following MIPS instructions places the value 0000 0000 0000 0000 0011 1111 0000 0000 in register \$t0?

- ☐ and \$t1, \$t2, \$t0
 ☐ add \$t1, \$t2, \$t0
☒ or \$t0, \$t1, \$t2
 ☐ add \$t0, \$t1, \$t2

Question 3. Which of the following MIPS codes loads the 32-bit constant 0000 0000 0010 0000 0000 0001 0000 0000 into register \$s0?

- ☐ lui \$s0, 256
 addi \$s0, \$s0, 32
 ☐ lui \$s0, 46
 addi \$s0, \$s0, 256
☒ lui \$s0, 32
 addi \$s0, \$s0, 256
 ☐ lw \$s0, 256
 add \$s0, \$s0, 32

Question 4. Suppose that the computer A, which has a 500MHz clock, runs a program P in 10 seconds. The computer B, which has twice the clock rate of A, runs P in 6 seconds. What is the number of clock cycles required for P on B?

- ☐ 3600×10^6 cycles
 ☐ 4600×10^6 cycles
☐ cycles
 ☒ 6000×10^6 cycles

Question 5. What is the machine instruction corresponding to the MIPS instruction: add \$s0,\$a0,\$t7?

- ☐ 0x00BF8020
 ☒ 0x008F8020
☐ 0x00AE8020
 ☐ 0x00AD8020

Question 6. Given a program P. Suppose that we have two computers A and B implementing the same instruction set architecture. Computer A has a clock cycle time of 1 ns and a CPI of 2.0 for P, and computer B has a clock cycle time of 2.5 ns and a CPI of 1.2 for P. How much faster is computer A than computer B for this program?

- ☒ 1.5 times
 ☐ times
☐ 1.3 times
 ☐ 0.9 times

Constructed-response questions

Question 7. Add comments to the following MIPS code and describe in one sentence what it computes. Assume that \$a0 is initially contains 8 and \$v0 is used for the output.

```
begin:  addi $t0, $zero, 0      .....
        addi $t1, $zero, 1      .....
loop:   slt  $t2, $a0, $t1      .....
        bne $t2, $zero, fin     .....
        add $t0, $t0, $t1      .....
        addi $t1, $t1, 2        .....
        j   loop               .....
fin:    add  $v0, $t0, $zero     .....

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```

Question 8. What are the MIPS instructions corresponding to the following machine instructions?

```
1000 1101 0010 1000 0000 0100 1011 0000 .....
0000 0010 0100 1000 0100 0000 0010 0000 .....
1010 1101 0010 1000 0000 0100 1011 0000 .....
```

Question 9. Compile the following C program into MIPS assembly code.

```
int func (int n)
{
    int i;
    for (i=0;i<n;i=i+1)
        if (i=n) return i;
}
```

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```

Question 10. Construct the full truth table described by the following Verilog module.

```
module func {A, B, C, S, D};
    input A, B, C;
    output S, D;
    assign S=A^B^C;
    assign D=(A&B) | (B&C) | (C&A);
endmodule
```

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